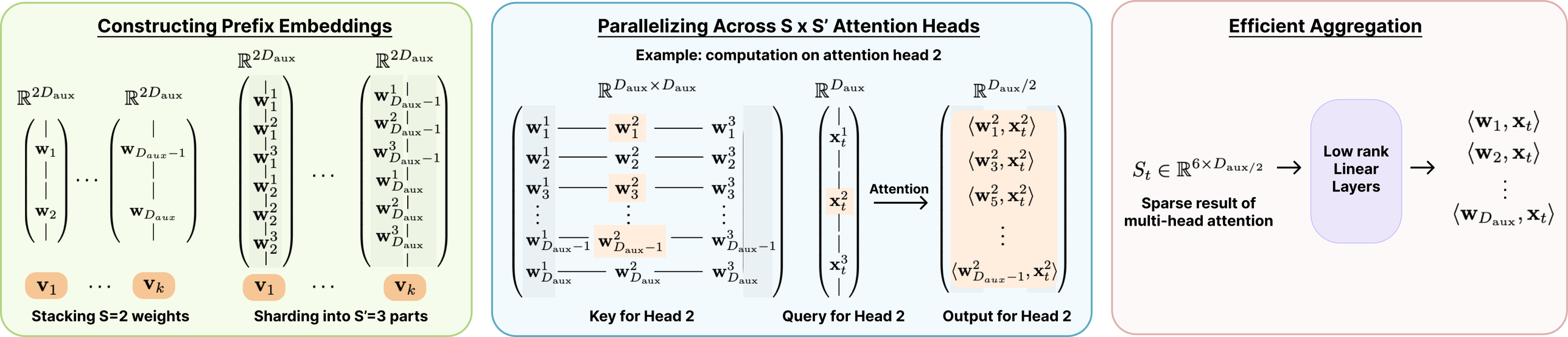
Constructing Prefix Embeddings
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbb{R}^{2D_{\text{aux}}}$
$\mathbf{w}_1$
$\mathbf{w}_2$
$\mathbf{w}_{D_{aux}-1}$
$\mathbf{w}_{D_{aux}}$
$\mathbf{v}_1$
$\mathbf{v}_k$
$\mathbf{v}_1$
$\mathbf{v}_k$
Stacking S=2 weights
Sharding into S'=3 parts
Parallelizing Across S x S' Attention Heads
Example: computation on attention head 2
$\mathbb{R}^{D_{\text{aux}} \times D_{\text{aux}}}$
$\mathbb{R}^{D_{\text{aux}}}$
$\mathbb{R}^{D_{\text{aux}}/2}$
Attention
$\langle \mathbf{w}_1^2, \mathbf{x}_t^2 \rangle$
$\langle \mathbf{w}_3^2, \mathbf{x}_t^2 \rangle$
$\langle \mathbf{w}_5^2, \mathbf{x}_t^2 \rangle$
$\langle \mathbf{w}_{D_{aux}-1}^2, \mathbf{x}_t^2 \rangle$
Key for Head 2
Query for Head 2
Output for Head 2
Efficient Aggregation
$S_t \in \mathbb{R}^{6 \times D_{\text{aux}}/2}$
Sparse result of multi-head attention
Low rank Linear Layers
$\langle \mathbf{w}_1, \mathbf{x}_t \rangle$
$\langle \mathbf{w}_2, \mathbf{x}_t \rangle$
$\langle \mathbf{w}_{D_{\text{aux}}}, \mathbf{x}_t \rangle$